Supplementary Appendix
Pixel Accuracy Is Not Biomarker Validity: RMDB–BiRAP Across Domains

Transparent negative results, calibration analyses, and confirmation preflight

Supplementary Methods

All after optimization CIMT analyses used only artifacts/manifest.csv split=train (2,603 patients). The fixed CIMT test set was not read. mBRSET analyses were grouped at patient level, and the 224 pixel mirror is interpreted as an external mechanism cohort rather than independent high resolution clinical confirmation.

The quality gate was defined from outcome blind CIMT development phenotype completeness and quantiles before Fundus-AVSeg ground truth was joined. Nested mBRSET calibration used a Platt model cross fitted only within the outer training data. These analyses were secondary and transparent; no new outcome guided feature search was undertaken.

Supplementary Tables

Table S1. CIMT patient level model results and the constrained by repeatability negative experiment.

Model / analysis	n	MAE (mm)	AUROC	Paired comparison	Interpretation
E10A conditional centile	2,603	0.10775	0.5550	vs E10_0: −0.00204 [−0.00282, −0.00124]	development reference
E12B gated by ICC core	2,603	0.10826	0.5446	vs E10A: +0.00057 [+0.00002, +0.00110]	worse; 6/25 not inferior

All values are repeated five seed × 5 fold outer cross validation summaries. Lower MAE is better.

Table S2. mBRSET calibrated reliable core increment over clinical covariates.

Analysis set	n (positive)	ΔAUROC	ΔAUPRC	ΔBrier	Evidence status
Overall	1,276 (912)	+0.00188 [-0.00588, +0.00942]	-0.00104 [-0.00714, +0.00526]	+0.00049 [-0.00064, +0.00165]	not supported
No diabetic retinopathy	882 (628)	-0.00052 [-0.01021, +0.00909]	-0.00026 [-0.00735, +0.00683]	+0.00062 [-0.00087, +0.00214]	not supported
Strict disc QC	1,105 (782)	+0.00558 [-0.00327, +0.01449]	+0.00152 [-0.00723, +0.00980]	-0.00017 [-0.00165, +0.00135]	not supported

R2 − R0; 95% paired bootstrap intervals. Lower Brier is better. All intervals for the primary increment cross zero.

Table S3. Negative, failed, and transport boundary experiments.

Experiment	Result	Conclusion
Between head disagreement	error Spearman −0.032; large error AUROC 0.475	not a valid uncertainty proxy
Fundus-AVSeg blind QC gate	100/100 images passed	no external QC separation; branch stopped
Quality overlap transport	domain AUROC: mBRSET 1.000; RAV 0.923; Fundus-AVSeg 0.921	cohorts not exchangeable; no pooling
Patient label permutation	R0 observed 0.6648; R2 observed 0.6663; p=0.005	pipeline integrity, not clinical increment

Table S4. Frozen external confirmation preflight.

Required item	Local status	Implication
Independent high resolution images	absent	cannot execute frozen confirmation
Linked patient level outcome	absent	RAV/Fundus-AVSeg are measurement only
MESA manifest + images	BLOCKED_EXTERNAL_DATA	controlled access data not present locally
Permitted confirmation contract	patient_id, image_id, eye, image_path, age, sex, cimt_mm	run frozen E10A once without retuning

Supplementary Figures

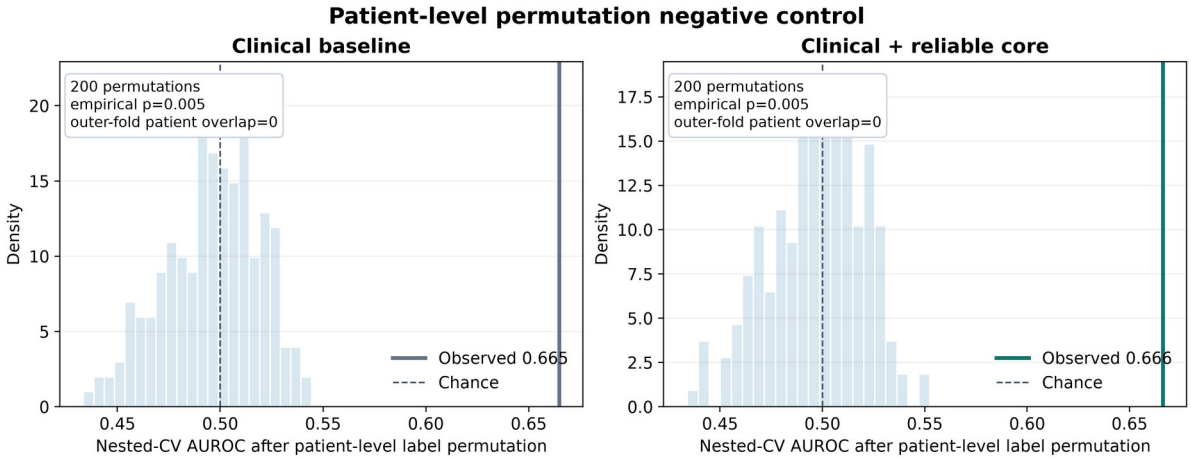


Figure S1. Patient level label permutation negative control. The observed out of fold AUROC values for R0 and R2 exceed their random label null distributions, while the R2–R0 increment remains clinically unsupported.

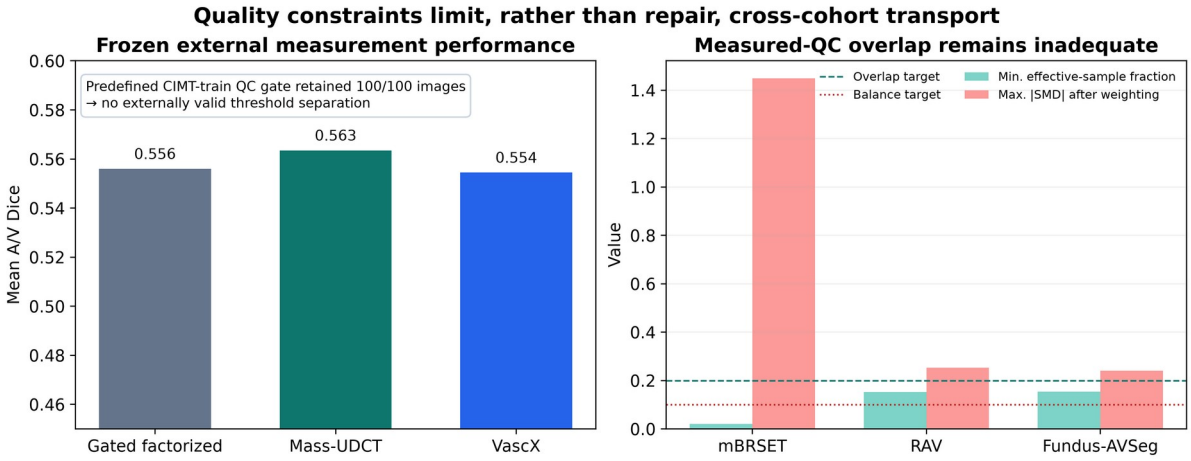


Figure S2. Outcome blind quality gate and phenotype transport audit. The fixed gate did not separate Fundus-AVSeg images, and covariate/quality transport overlap was insufficient for pooled outcome inference.

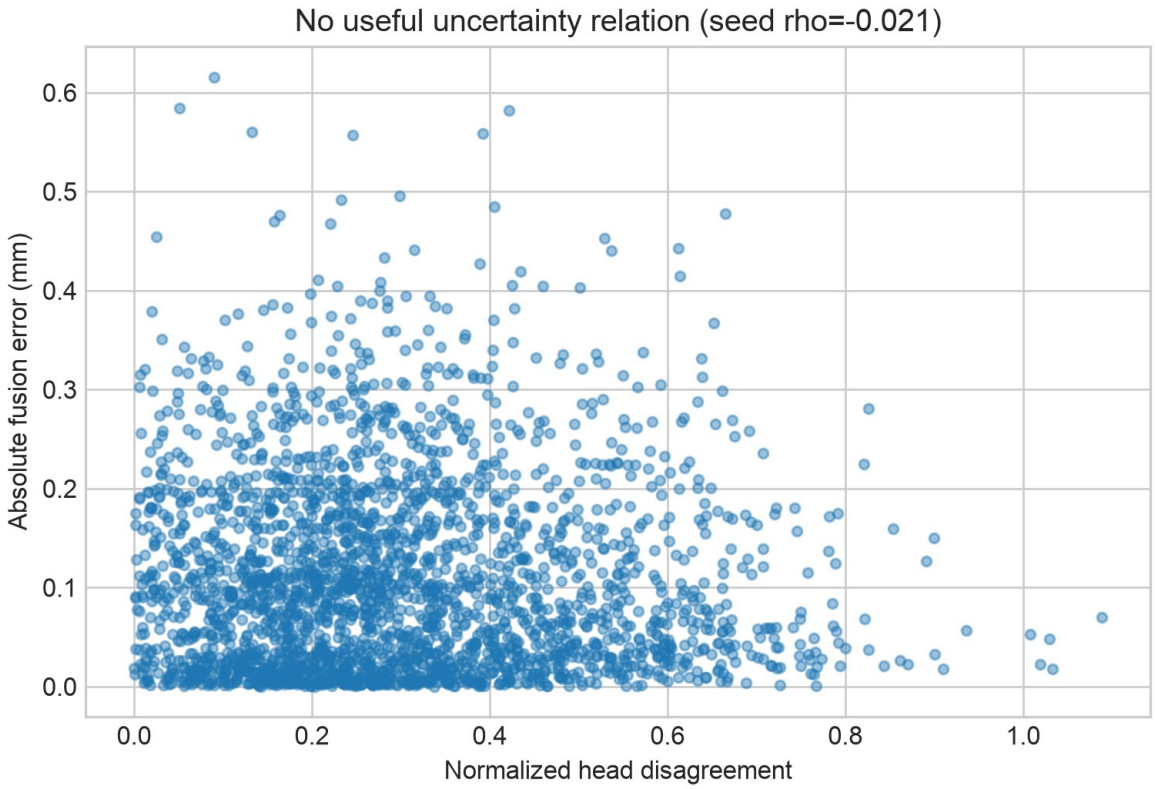


Figure S3. Rejected uncertainty hypothesis. Between head disagreement did not track absolute prediction error and was removed from the uncertainty claim.

Submission readiness items that remain dependent on the authors

Before journal submission, insert author names and affiliations; institutional review board and consent language; data use statements for each source; and verified bibliographic citations for FIVES, RAV, Fundus-AVSeg, mBRSET, and China Fundus–CIMT. These author/governance fields are intentionally not fabricated in this draft.

Supplementary transport and access audits

The following analyses were outcome blind and used only the frozen four phenotype reliable core. CIMT input was restricted to split=train; no CIMT validation or test row and no systemic outcome was read. Repeated views were collapsed to the patient and eye median before domain classification.

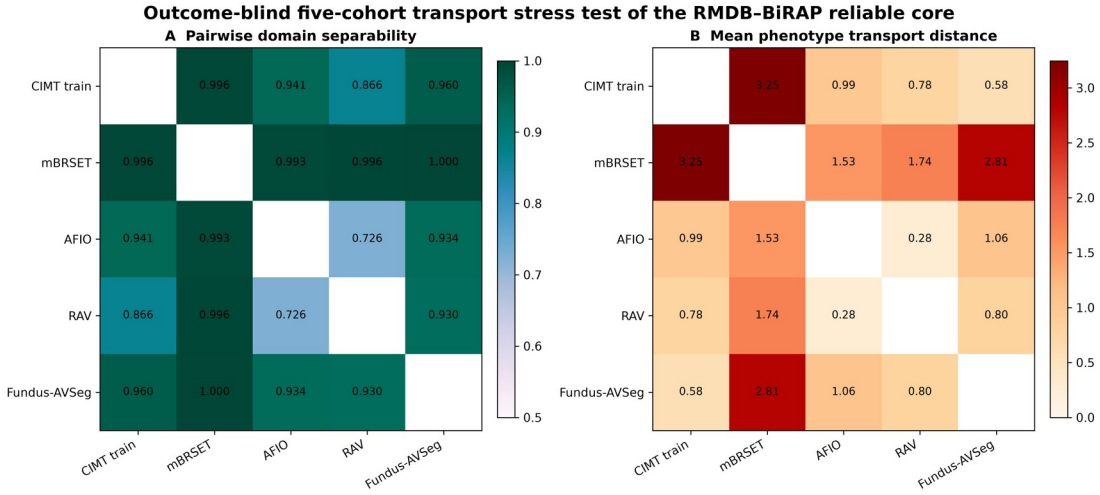


Figure S4. Exact five cohort transport audit. Pairwise grouped by patient domain AUROC and normalized Wasserstein distance show that the four retained phenotype scales remain not exchangeable across image resources.

Table S5. All pairwise five cohort transport results.

Pair	n eyes	Domain AUROC	95% CI	Mean normalized W	Mean |ΔSpearman|
CIMT train vs mBRSET	7729	0.9958	0.9942–0.9973	3.2467	0.3325
CIMT train vs AFIO	5306	0.9412	0.9209–0.9581	0.9911	0.1319
CIMT train vs RAV	5412	0.8664	0.8382–0.8923	0.7814	0.1143
CIMT train vs Fundus-AVSeg	5306	0.9605	0.9412–0.9766	0.5754	0.0764
mBRSET vs AFIO	2623	0.9931	0.9789–0.9999	1.5313	0.2077
mBRSET vs RAV	2729	0.9962	0.9914–0.9997	1.7388	0.4463
mBRSET vs Fundus-AVSeg	2623	1.0000	1.0000–1.0000	2.8095	0.4082
AFIO vs RAV	306	0.7257	0.6690–0.7811	0.2814	0.2432
AFIO vs Fundus-AVSeg	200	0.9341	0.8980–0.9631	1.0571	0.2083
RAV vs Fundus-AVSeg	306	0.9300	0.9026–0.9579	0.7997	0.0952

Table S6. Frozen structure migration and public pixel feasibility results.

Resource	Analysis	n	Result	Decision
FIVES held out	Frozen binary vessel migration	200	Vessel Dice 0.356	Negative domain shift; no retuning
STARE AH	Frozen binary vessel migration	20	Vessel Dice 0.167	Negative domain shift; no retuning
AFIO	Frozen extraction completeness	100	100 vessel pairs; 99 A/V pairs	One absent source pair left missing
NHANES	Public retinal pixel feasibility	2,934/3,863 records	Grading/clinical files found; raw retinal pixels unavailable	Not an image cohort